



BAAZ JKBOSE

🔥 2026 CLASS 11th 🔥

Cell:
The Fundamental Unit of Life

BIOLOGY

LECTURE-1

BY-DAVOOD SIR



TOPICS TO BE COVERED

- 1 Introduction to Cell
- 2 Cell theory
- 3 Size of cells.
- 4 Cell diversity and division of labour
- 5 Structure of a cell.

Science

Physics

Chemistry

Biology

Bio = livings
logy = study of

Father: Aristotle

Arshid Sir

Botany

→ Study of plants
→ Father: Theophrastus

Zoology

→ Study of Animals
→ Aristotle

(EO of N.G.)

Chapters of Class 11th Zoology

1 Cell, The Unit of Life ✓

5 MCQs - Neet 2025. (5x4=20 marks)

2 Cell Cycle & Cell division. ✓

3 Biomolecules ✓

4 Structural organisation in Animals ✓

5 Animal Kingdom ✓

6 Breathing & Exchange of Gases

7 Body Fluids & Circulation

8 Excretory Products & Their Elimination.

9 Locomotion & Movement.

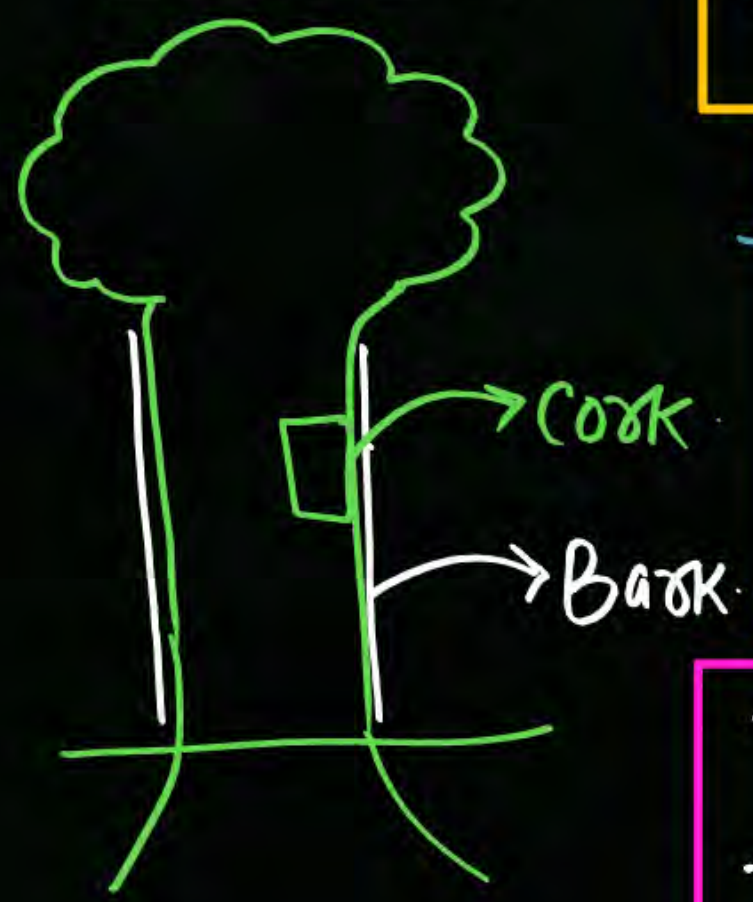
10 Neural Control & Coordination.

11 Chemical Coordination & Integration.

Human physiology

Discovery of a Cell

Robert Hooke
1665
Oak tree



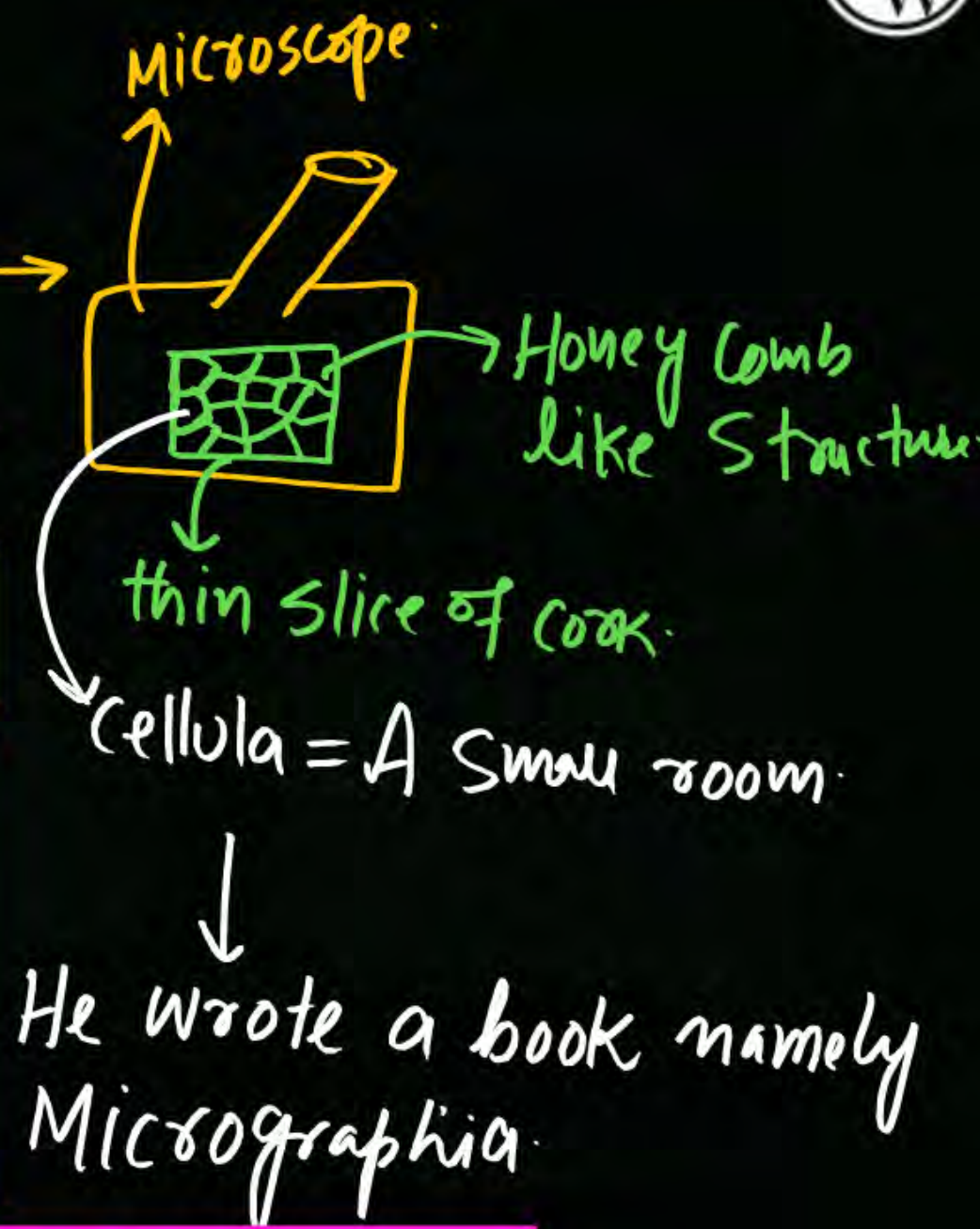
→ peice of cork → thin slices of cork



→ Robert Hooke discovered a dead cell.

→ Anton-van-Leewenhoek → Discovered a living cell
↓
Bacterium

→ Unicellular organism: living made up of single cell.
e.g → Amoeba.
→ Multicellular organism: living made of many cells.
↓
plants, Animals.



Introduction to Cell

- All living organisms are made up of one cell (unicellular) some are made up of more cells (multicellular). They are the smallest unit of life that can carry out all the functions necessary for life.
- **Anton Von Leeuwenhoek** described a living cell.
- The invention of the microscope and its improvement leading to the electron microscope revealed all the structural details of the cell.

Cell Theory

1839

- Schleiden → Botanist: All the plants are made of cells
 - Schwann → Zoologist: All the Animals are made of cells
- All the livings are made of cells.
- Incomplete cell theory.
-
- Rudolf Virchow → Finalized cell theory by modifying previous cell theory
 - All the livings are made of cells.
 - Omnis cellula-e-cellula (New cells arise from pre-existing cells)

↓
theory of cell lineage.

Cell Theory

Cell theory is a fundamental concept in biology that describes the properties and functions of cells, the basic unit of life. It was developed in the 19th century by scientists Matthias Schleiden (Botanist) & Theodor Schwann (Zoologist).

Cell theory is as: All living organisms are made up of cells.

{ Schwann reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells.

Cell Theory

All cells arise from pre-existing cells (**Omnis cellula-e cellula**): This principle, proposed by Rudolf Virchow, states that new cells are formed by the division of existing cells and is called as Theory of cell lineage.

Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape. Cell theory as understood today is:

- i. All living organisms are composed of cells and products of cells.
- ii. All New cells arise from pre-existing cells.

Cell size

→ pleuro pneumonia like organism (PPLO)

- Mycoplasmas, the smallest cells, are only 0.3 μm in length. ✓
- The largest isolated single cell is the egg of an ostrich.
- Among multicellular organisms, human RBC's are about 7.0 μm in diameter.
- Nerve cells are some of the longest cells.

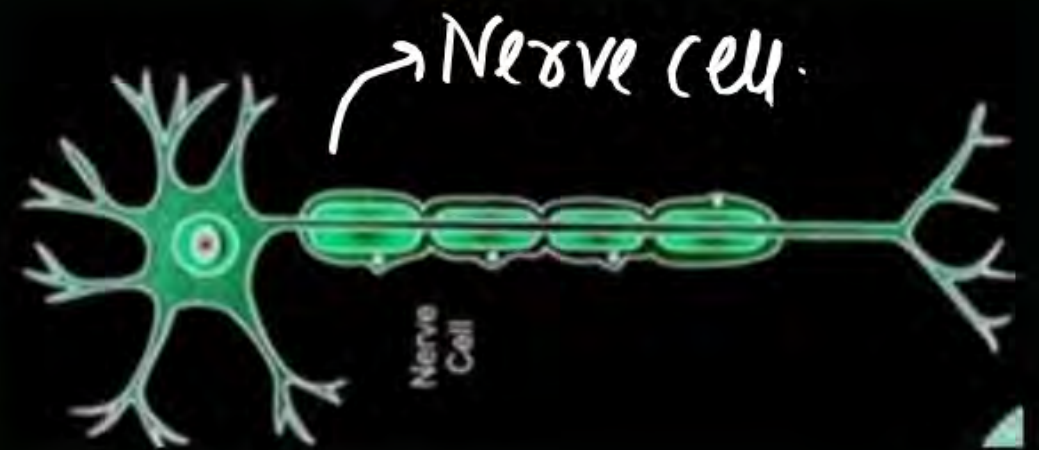
→ In Humans, sperm is the smallest cell.

→ In Humans, female egg cell is the largest cell.

→ ostrich egg.



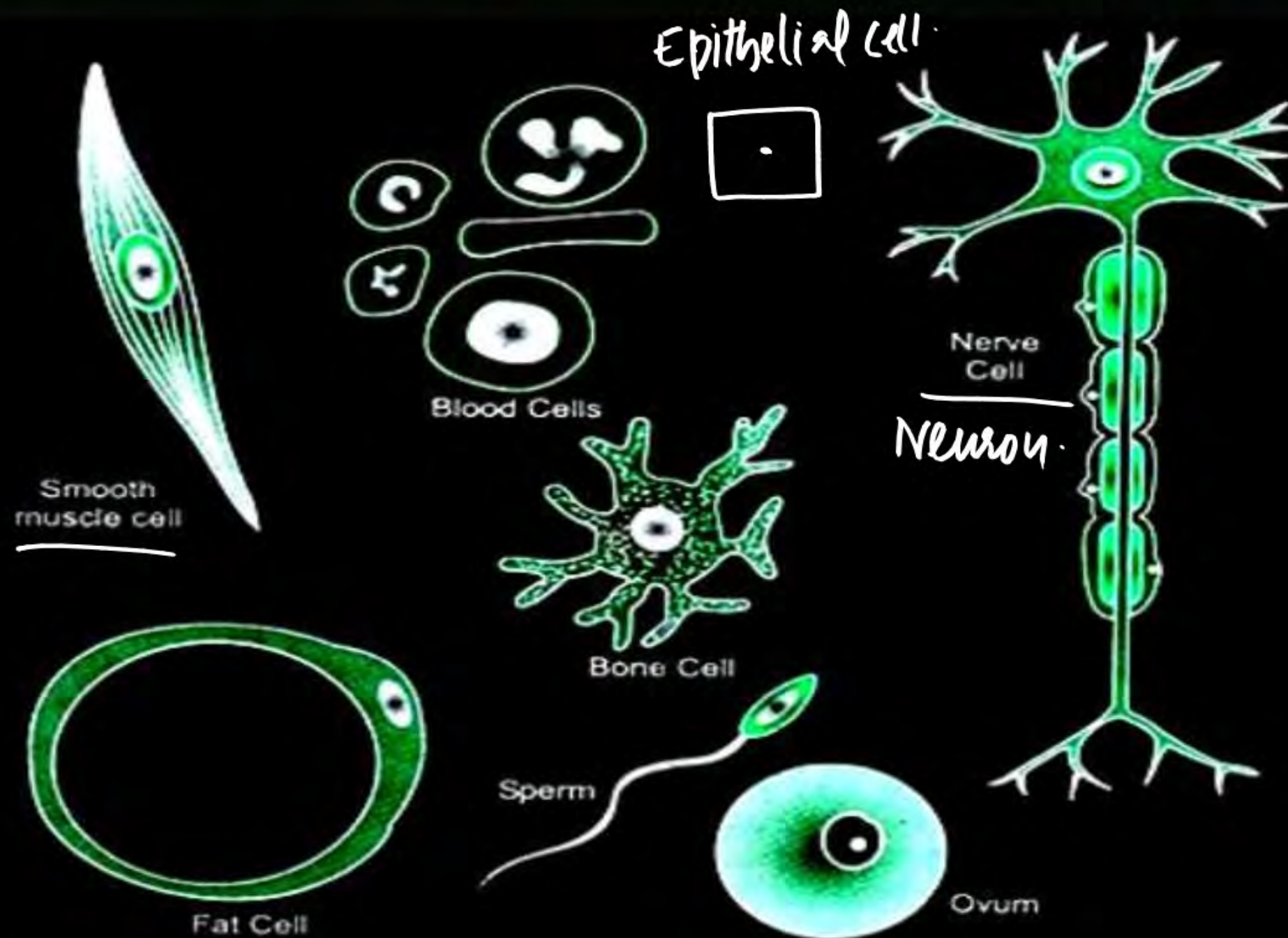
→ RBC.



→ Nerve cell.

Nerve Cell

Cell Diversity & Division of Labor



Cell Diversity & Division of Labor

1. Cell diversity is the variety of cell types that make up tissues and organisms. The human body has around 200 types of cells, They may be disc-like, polygonal, columnar, cuboid, thread like, or even irregular. The shape of the cell may vary with the function they perform.
2. In biology, **division of labor (DOL)**. It refers to the distribution of different functions among different parts or different cells in an organism. DOL can be seen generally in multicellular organisms.

Question: NEET-1993

Kon Banayga Nabkaar (KBN)

Names of Schleiden and Schwann are associated with:

- 1 Protoplasm as the physical basis of life
- 2 Cell theory. ✓
- 3 Theory of cell lineage
- 4 Nucleus functions as control centre of cell



Question: NEET-2019

Kon Banayga Nabkaar (KBN)

The concept of "Omnis cellula-e-cellula" regarding cell division was first proposed by

- 1 Aristotle
- 2 Rudolf Virchow.
- 3 Theodore Schwann
- 4 Schleiden



Thank
YOU